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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/439,483	09/15/2021	Young Jae CHOI	HI-1864	2949

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KED & ASSOCIATES, LLP
P.O. Box 8638
Reston, VA 20195

EXAMINER
CARTER, WILLIAM JOSEPH

ART UNIT	PAPER NUMBER
2875	

NOTIFICATION DATE	DELIVERY MODE
08/30/2024	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

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Office Action Summary	Application No. 17/439,483	Applicant(s) CHOI et al.	
	Examiner WILLIAM J CARTER	Art Unit 2875	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 23 May 2024.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☑ This action is **FINAL.** 2b) ☐ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-20 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☑ Claim(s) 1 and 4-10 is/are allowed.

7) ☑ Claim(s) 2-3 and 11-20 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 15 September 2021 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☑ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☑ All b)☐ Some** c)☐ None of the:

1.☑ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date 4/19/24.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Claim Rejections - 35 USC § 103

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

Claims 2-3 and 11-20 are rejected under 35 U.S.C. 103 as being unpatentable over Ichihara (2007/0030676) in view of Chosa et al. (2007/0109792).

With respect to claim 2, Ichihara teaches a lighting device (Fig. 5) comprising: a substrate (1); a light source (2) disposed on the substrate, the light source including a plurality of light emitting devices (Fig. 5); a resin layer (3) disposed on the substrate and the light source (Fig. 5); and a first reflective layer (52) disposed on the resin layer (Fig. 5), wherein the resin layer includes an exit surface (top surface of 3) facing the light source, wherein the light source includes a plurality of light emitting devices (Fig. 5), wherein the exit surface of the resin layer includes a plurality of convex portions (31) facing to each of the plurality of light emitting devices (Fig. 5), wherein each of the plurality of light emitting devices is disposed on a virtual curve, wherein a difference in

diameter of each of virtual circles formed by each of the plurality of convex portions is within 10% (Fig. 5).

Ichihara does not explicitly teach a first reflective layer disposed on an upper surface of the resin layer, wherein the resin layer is disposed between the lower surface of the first reflective layer and an upper surface of each of the plurality of light emitting devices and between an upper surface of the substrate and the lower surface of the first reflective layer (claim 2).

As for claim 2, Chosa also drawn to lighting devices, teaches a first reflective layer (127) disposed on an upper surface of the resin layer (121), wherein the resin layer is disposed between the lower surface of the first reflective layer and an upper surface of each of the plurality of light emitting devices (124) and between an upper surface of the substrate (123) and the lower surface of the first reflective layer (Figs. 6-7).

It would have been obvious to one of ordinary skill in the art, before the effective filing date of the present invention, to use the arrangement of Chosa in the lighting device of Ichihara, in order to avoid light loss and improve the luminance of the lighting device (paragraph 5 of Chosa).

As for claim 3, Ichihara further teaches a lighting device (Fig. 5) comprising: a substrate (1); a light source (2) disposed on the substrate; a resin layer (3) disposed on the substrate and the light source; and a first reflective layer (52) disposed on the resin layer, wherein the resin layer includes an exit surface (top surface of 3) facing the light source, wherein the light source includes a plurality of light emitting devices (Fig. 5),

wherein the exit surface of the resin layer includes a plurality of convex portions (31) facing to each of the plurality of light emitting devices, wherein each of the plurality of light emitting devices is disposed on a virtual curve (Fig. 5), wherein a plurality of straight lines passes through each of the plurality of light emitting devices and center of virtual circles formed by each of the plurality of convex portions, and wherein a first angle between the each of plurality of straight lines and each tangent line at a point where each of the plurality of light emitting devices is in contact with the virtual curve increases in one direction (Fig. 5).

With respect to claim 11, the embodiment shown in Fig. 5 teaches all of the claimed elements, as is discussed above, except for explicitly teaching wherein a maximum distance from a center of each of the light emitting devices to each of the convex portions is equal to each other, wherein each of the plurality of light emitting devices is disposed on a circumference formed by a corresponding one of the virtual circle, wherein a plurality of straight lines passing through centers of each of the plurality of light emitting devices and a center of each of the virtual circles are parallel to each other (claim 11).

As for claim 11, in the embodiment shown in Fig. 7, Ichihara teaches wherein a maximum distance from a center of each of the light emitting devices to each of the convex portions is equal to each other, wherein each of the plurality of light emitting devices is disposed on a circumference formed by a corresponding one of the virtual circles, wherein a plurality of straight lines passing through centers of each of the

plurality of light emitting devices and a center of each of the virtual circles are parallel to each other (Fig. 7).

It would have been obvious to one of ordinary skill in the art, before the effective filing date of the present invention, to use the alignments taught in Fig. 7 of Ichihara in the embodiment shown in Fig. 5 of Ichihara, in order to improve the quality of a lighting fixture (paragraph 55 of Ichihara).

As for claim 12, Ichihara further teaches wherein the plurality of light emitting devices include first to third light emitting devices, wherein the plurality of convex portions includes first to third convex portions, wherein the plurality of straight lines includes first to third straight lines, wherein a plurality of tangent lines includes first to third tangent lines, wherein a first angle between the first straight line and the first tangent line is a first obtuse angle, wherein a first angle formed by the second straight line and the second tangent line is a second obtuse angle, wherein a first angle formed by the third straight line and the third tangent line is a third obtuse angle, wherein the second obtuse angle is greater than the first obtuse angle and smaller than the third obtuse angle (Fig. 5 and paragraph 46).

With respect to claim 13, the embodiment shown in Fig. 5 teaches all of the claimed elements, as is discussed above, except for explicitly teaching wherein a straight line connecting adjacent light emitting devices has a slope with respect to a horizontal straight line (claim 13).

As for claim 13, in the embodiment shown in Fig. 7, Ichihara teaches wherein a straight line connecting adjacent light emitting devices has a slope with respect to a horizontal straight line (Fig. 7).

It would have been obvious to one of ordinary skill in the art, before the effective filing date of the present invention, to use the alignments taught in Fig. 7 of Ichihara in the embodiment shown in Fig. 5 of Ichihara, in order to improve the quality of a lighting fixture (paragraph 55 of Ichihara).

As for claim 14, Ichihara teaches wherein curvatures of virtual circles formed by each of the plurality of convex portions are equal to each other, and wherein the plurality of convex portions includes a hemispherical, elliptical, or aspherical shape (Fig. 5).

As for claim 15, Ichihara teaches wherein a diameter of a circle formed by one of the convex portions is greater than a thickness of the resin layer (Fig. 5).

As for claim 16, Ichihara teaches wherein a diameter of a circle formed by one of the convex portions is greater than a thickness of the resin layer (Fig. 5).

With respect to claim 17, although Ichihara does not explicitly teach comprising a second reflective layer disposed between the resin layer and the substrate such that the resin layer is between the first reflective layer and the second reflective layer, wherein a thickness of the resin layer is less than a distance from a rear surface of the resin layer to the exit surface of the resin layer; Choa teaches a second reflective layer (120) disposed between the resin layer and the substrate such that the resin layer is between

the first reflective layer and the second reflective layer (Fig. 7), wherein a thickness of the resin layer is less than a distance from a rear surface of the resin layer to the exit surface of the resin layer (Fig. 7). It would have been obvious to one of ordinary skill in the art, before the effective filing date of the present invention, to use the arrangement of Chosa in the lighting device of Ichihara, in order to avoid light loss and improve the luminance of the lighting device (paragraph 5 of Chosa).

With respect to claim 18, the embodiment shown in Fig. 5 teaches all of the claimed elements, as is discussed above, except for explicitly teaching wherein a maximum distance from a center of each of the light emitting devices to each of the convex portions is equal to each other, wherein each of the plurality of light emitting devices is disposed on a circumference formed by a corresponding one of the virtual circles, wherein a plurality of straight lines passing through centers of each of the plurality of light emitting devices and a center of each of the virtual circles are parallel to each other (claim 18).

As for claim 18, in the embodiment shown in Fig. 7, Ichihara teaches wherein a maximum distance from a center of each of the light emitting devices to each of the convex portions is equal to each other, wherein each of the plurality of light emitting devices is disposed on a circumference formed by a corresponding one of the virtual circles, wherein a plurality of straight lines passing through centers of each of the plurality of light emitting devices and a center of each of the virtual circles are parallel to each other (Fig. 7).

It would have been obvious to one of ordinary skill in the art, before the effective filing date of the present invention, to use the alignments taught in Fig. 7 of Ichihara in the embodiment shown in Fig. 5 of Ichihara, in order to improve the quality of a lighting fixture (paragraph 55 of Ichihara).

As for claim 19, Ichihara teaches wherein the plurality of light emitting devices include first to third light emitting devices, wherein the plurality of convex portions includes first to third convex portions, wherein the plurality of straight lines includes first to third straight lines, wherein a plurality of tangent lines includes first to third tangent lines, wherein a first angle between the first straight line and the first tangent line is a first obtuse angle, wherein a first angle formed by the second straight line and the second tangent line is a second obtuse angle, wherein a first angle formed by the third straight line and the third tangent line is a third obtuse angle, wherein the second obtuse angle is greater than the first obtuse angle and smaller than the third obtuse angle (Fig. 5 and paragraph 46).

With respect to claim 20, the embodiment shown in Fig. 5 teaches all of the claimed elements, as is discussed above, except for explicitly teaching wherein a straight line connecting adjacent light emitting devices has a slope with respect to a horizontal straight line (claim 20).

As for claim 20, in the embodiment shown in Fig. 7, Ichihara teaches wherein a straight line connecting adjacent light emitting devices has a slope with respect to a horizontal straight line (Fig. 7).

It would have been obvious to one of ordinary skill in the art, before the effective filing date of the present invention, to use the alignments taught in Fig. 7 of Ichihara in the embodiment shown in Fig. 5 of Ichihara, in order to improve the quality of a lighting fixture (paragraph 55 of Ichihara).

Allowable Subject Matter

Claims 1 and 4-10 are allowed.

With respect to claims 1 and 4-10, the prior art does not teach or suggest wherein the plurality of light sources is disposed on a virtual curve and include a first light source and a second light source adjacent to each other, wherein the plurality of convex portions includes a first convex portion facing the first light source and a second convex portion facing the second light source, wherein a first virtual straight line passing through a center of a circle formed by the first convex portion at a first point where the first light source is in contact with the virtual curve and a second virtual straight line passing through a center of a circle formed by the second convex portion at a second point where the second light source is in contact with the virtual curve is parallel to each other, wherein an angle between a first tangent line contacting the virtual curve at the first point and the first straight line includes a first obtuse angle; along with the other limiting elements of claim 1.

Response to Arguments

Applicant's arguments with respect to claims 2-3 and 11-20 have been considered but are moot because the new ground of rejection does not rely on any

reference applied in the prior rejection of record for any teaching or matter specifically challenged in the argument.

Conclusion

Applicant's amendment necessitated the new ground(s) of rejection presented in this Office action. Accordingly, **THIS ACTION IS MADE FINAL**. See MPEP § 706.07(a). Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

A shortened statutory period for reply to this final action is set to expire THREE MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date of this final action and the advisory action is not mailed until after the end of the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any extension fee pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the date of this final action.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to WILLIAM JOSEPH CARTER whose telephone number is (571)272-0959. The examiner can normally be reached M-F 8am-5pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, ABDULMAJEED AZIZ can be reached on 571-270-5046. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/WILLIAM J CARTER/
Primary Examiner, Art Unit 2875
8/27/2024